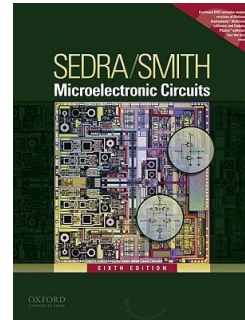


Partie 1 : Dispositifs et Circuits de base

- 5 : MOSFET

Transistor MOS



Chapter 5



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ELEC1530 – Partie 1 : MOSFET - 1

1

Introduction : Système de régulation de tension

Alimentation DC
(à partir d'une source de puissance AC)

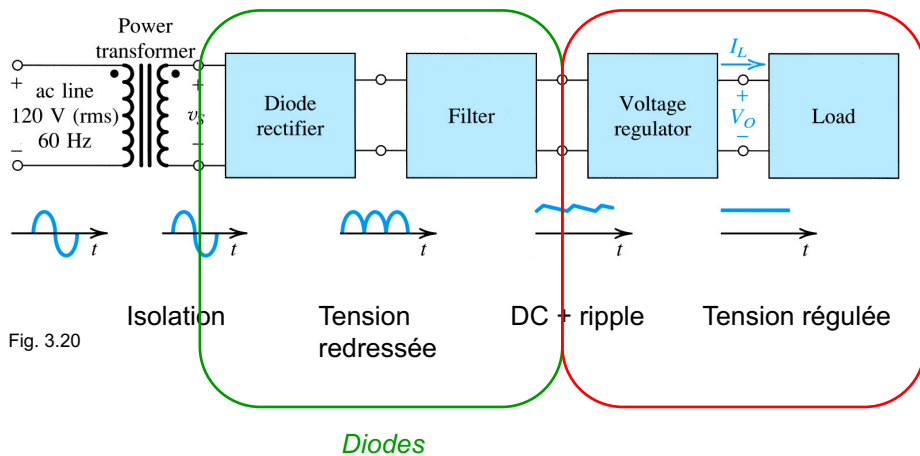


Fig. 3.20



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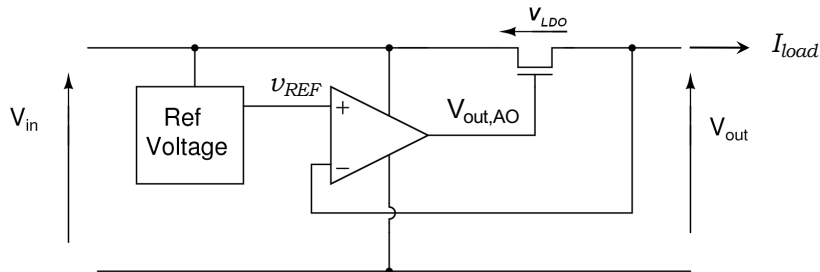
ELEC1530 – Partie 1 : MOSFET - 2

2

Régulateur LDO (Linear Drop Out)

LELEC2532

Tension de sortie = Tension de référence (fixe)



Utilisation d'un transistor MOS en résistance contrôlée

$$V_{OUT} = V_{IN} - R_{MOS} \cdot I_{load} \quad \text{ssi} \quad V_{IN} > V_{REF}$$

$$R_{MOS} = f(V_{out,AO}) \quad \text{telle que} \quad V_{OUT} = V_{REF}$$

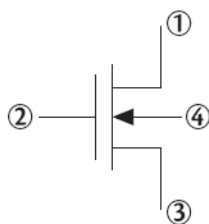


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3

SPICE avec MOSFET de la Sedra_Lib



```
MXXXXXXXX 1 2 3 4 mname <L=val> <W=val> <AD=val> <AS=val>
<PD=val> <PS=val> <NRD=val> <NRS=val> <OFF>
<IC=vds, vgs, vbs> <TEMP=t> <M=# of parallel
devices>
```

* Level-1 Model for the 0.5-um PMOS Transistor (Part PMOS0P5)

* (created by Anas Hamoui & Olivier Trescases)

```
.model PMOS0P5 PMOS(Level=1 VTO=-0.8 GAMMA=0.45 PHI=0.75
+ LD=0.09E-06 WD=0 UO=115 LAMBDA=0.2 TOX=9.5E-9 PB=0.9 CJ=0.93E-3
+ CJSW=170E-12 MJ=0.5 MJSW=0.35 CGDO=0.35E-9 JS=5E-9 CGBO=0.38E-9
+ CGSO=0.35E-9)
```

* Level-1 Model for the 0.5-um NMOS Transistor (Part NMOS0P5)

* (created by Anas Hamoui & Olivier Trescases)

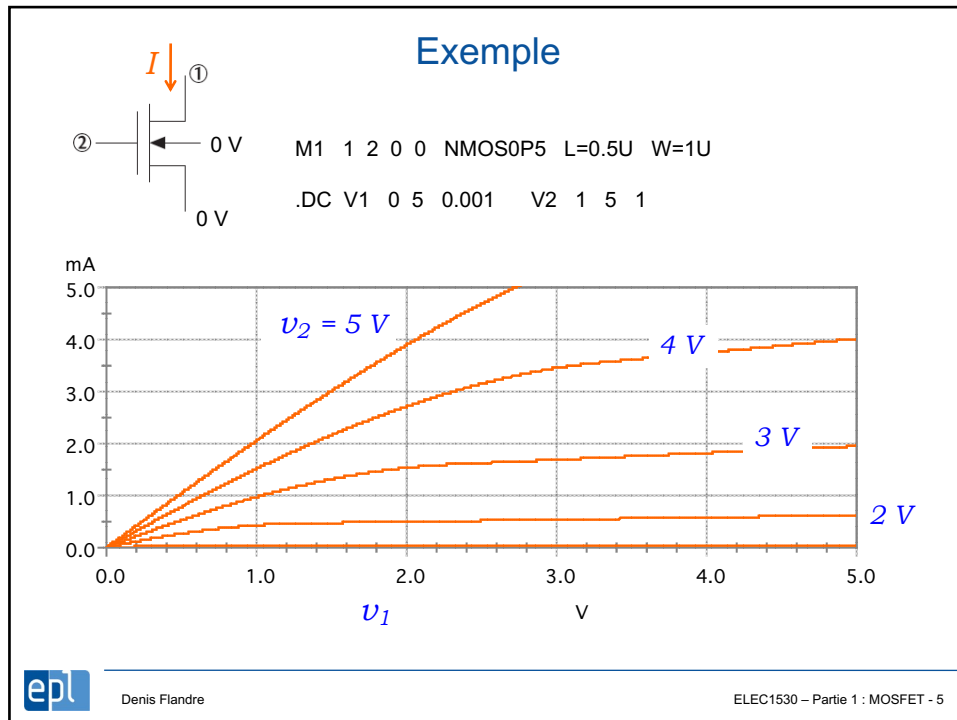
```
.model NMOS0P5 NMOS(Level=1 VTO=0.7 GAMMA=0.5 PHI=0.8
+ LD=0.08E-06 WD=0 UO=460 LAMBDA=0.1 TOX=9.5E-9 PB=0.9 CJ=0.57E-3
+ CJSW=120E-12 MJ=0.5 MJSW=0.4 CGDO=0.4E-9 JS=10E-9 CGBO=0.38E-9
+ CGSO=0.4E-9)
```



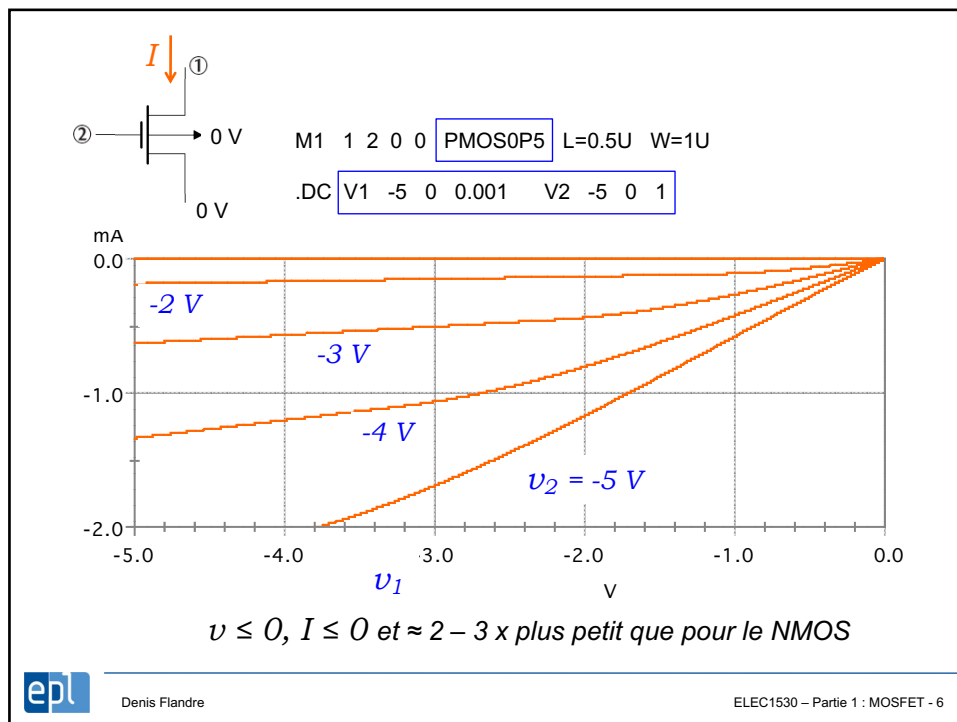
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4



5



6

Plan du cours

- 5.1 Structure du transistor MOS
- 5.2 Caractéristiques courant – tension
- 5.3 Circuits MOS DC
- 5.4 Amplificateurs MOS en 'grand-signal'
- 5.5 Fonctionnement et modèles 'petit-signal'
- 5.6 Configurations d'amplificateurs MOS
- 5.7 Polarisation
- 5.8 Amplificateurs MOS discrets
- 5.9 Effets supplémentaires

5.1 Structure du transistor MOS

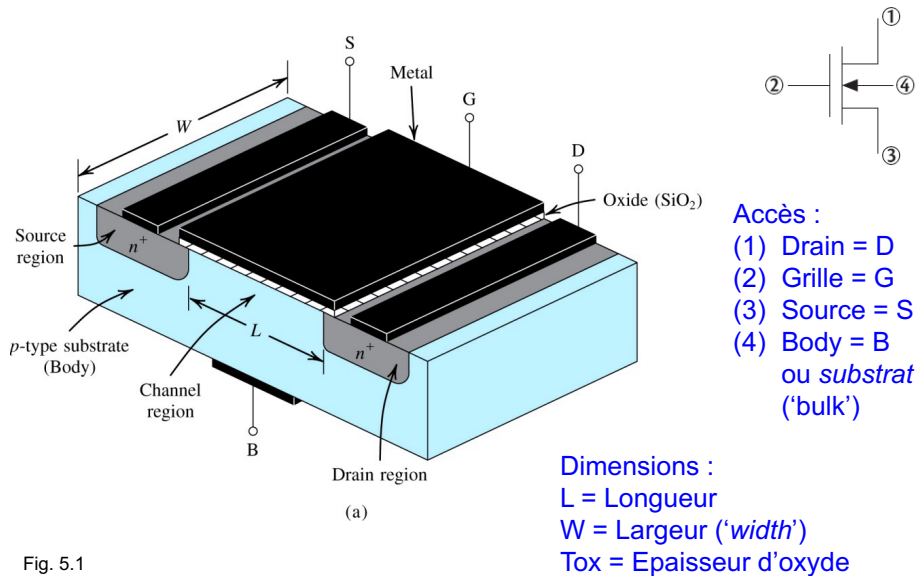
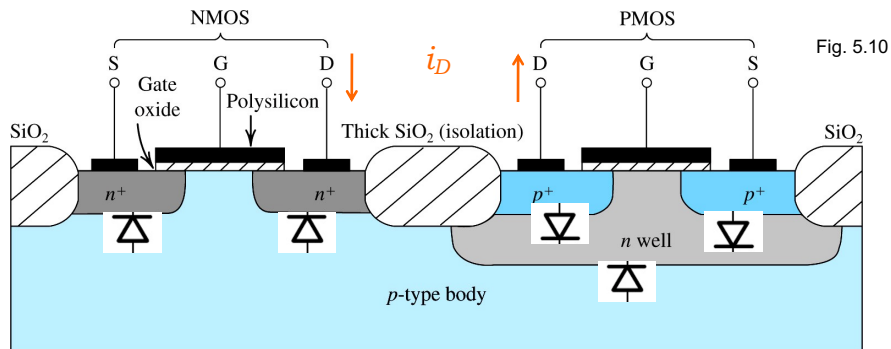


Fig. 5.1

Technologie MOS complémentaire ou CMOS



Paramètres électriques : Tension de seuil, V_{T0} et

Mobilité, U_0

• NMOS : i_D (entrant) $\gg 0$ si $v_{GS} > V_{tn}$ (> 0)

$i_D \div U_0$

• PMOS : i_D (sortant) $\gg 0$ si $v_{GS} < V_{tp}$ (< 0)

$U_{0\text{NMOS}} \approx 2,5 \cdot U_{0\text{PMOS}}$



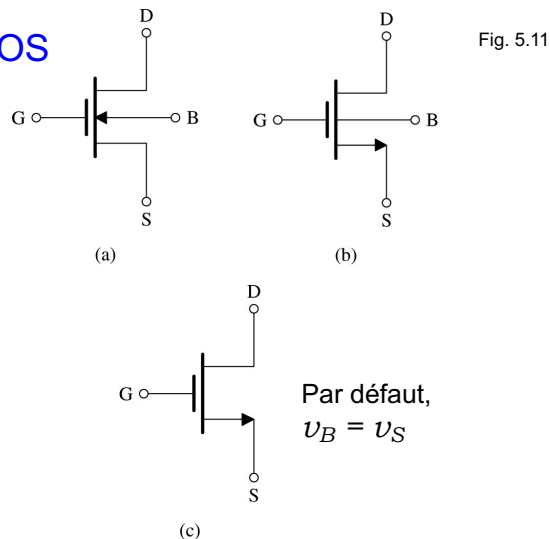
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5.2 Caractéristiques Courant - Tensions

Symboles du NMOS



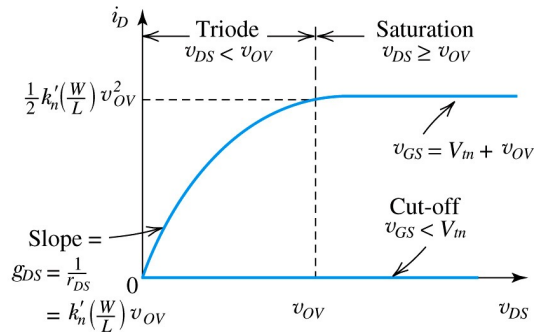
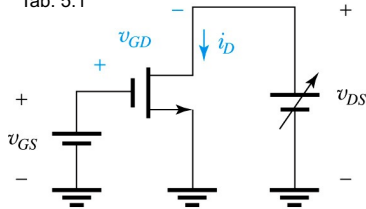
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Régions de fonctionnement du NMOS

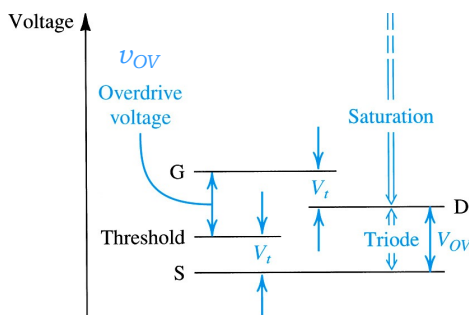
Tab. 5.1



- $v_{GS} < V_{tn} : i_D \approx 0$ (régime **coupé** ou 'cut-off')
- 'overdrive' $v_{OV} = v_{GS} - V_{tn} > 0 : i_D > 0$ (régime **passant** ou 'turned-on')
 - $v_{DS} < v_{OV} : i_D(v_{DS}, v_{OV})$ (régime **triode**)
 - $v_{DS} \geq v_{OV} : i_D(v_{OV})$ (régime **saturé**)

Equations I-V

Fig. 5.12



Process Transconductance
Parameter

$$k_n' = \mu_n C_{ox} = \mu_n \frac{\epsilon_{ox}}{t_{ox}}$$

$$\epsilon_{ox} = 3.9 \epsilon_0$$

$$k_n = k_n' \cdot W / L \text{ (ou } \beta \text{)}$$

Saturation Region :

$$v_{DS} \geq v_{OV} \Leftrightarrow v_{GD} \leq V_{tn}$$

→

Triode Region :

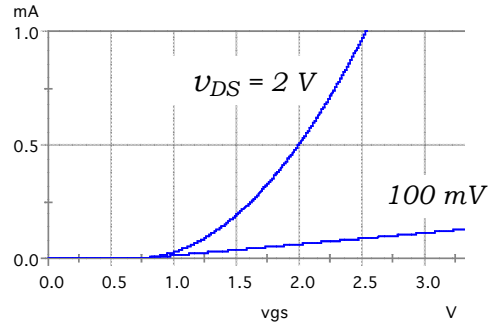
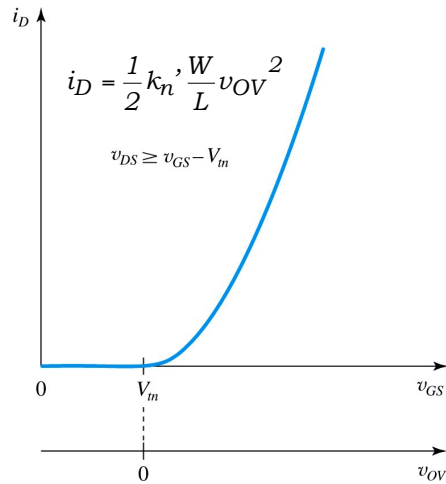
$$v_{DS} < v_{OV} \Leftrightarrow v_{GD} > V_{tn}$$

$$i_D = k_n' \frac{W}{L} \left[(v_{GS} - V_{tn}) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

$$\text{ou } i_D = k_n' \frac{W}{L} \left[v_{OV} - \frac{1}{2} v_{DS} \right] \cdot v_{DS}$$

Courbes I-V

• $i_D - v_{GS}$



si v_{DS} faible :

$$i_D = k_n' \frac{W}{L} \left[(v_{GS} - V_{tn}) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

$$\rightarrow k_n' \frac{W}{L} v_{OV} \cdot v_{DS} \quad (\text{linéaire})$$

Fig. 5.14

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• $i_D - v_{DS}$

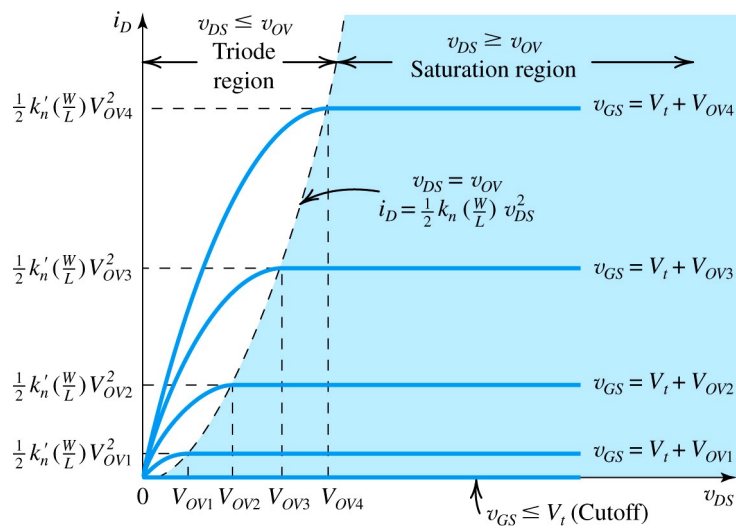
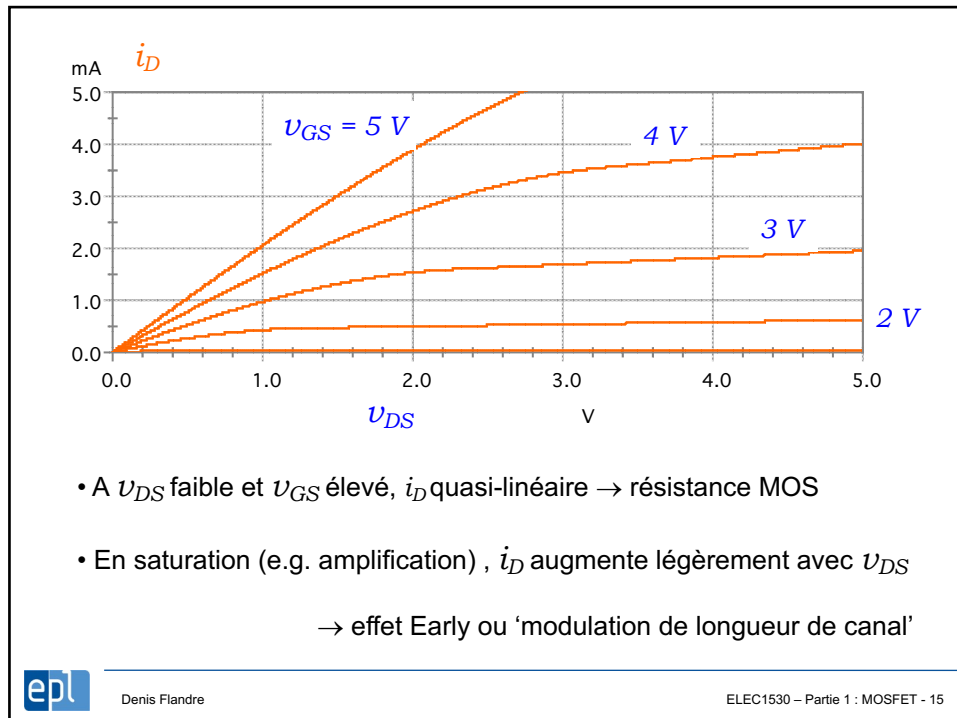
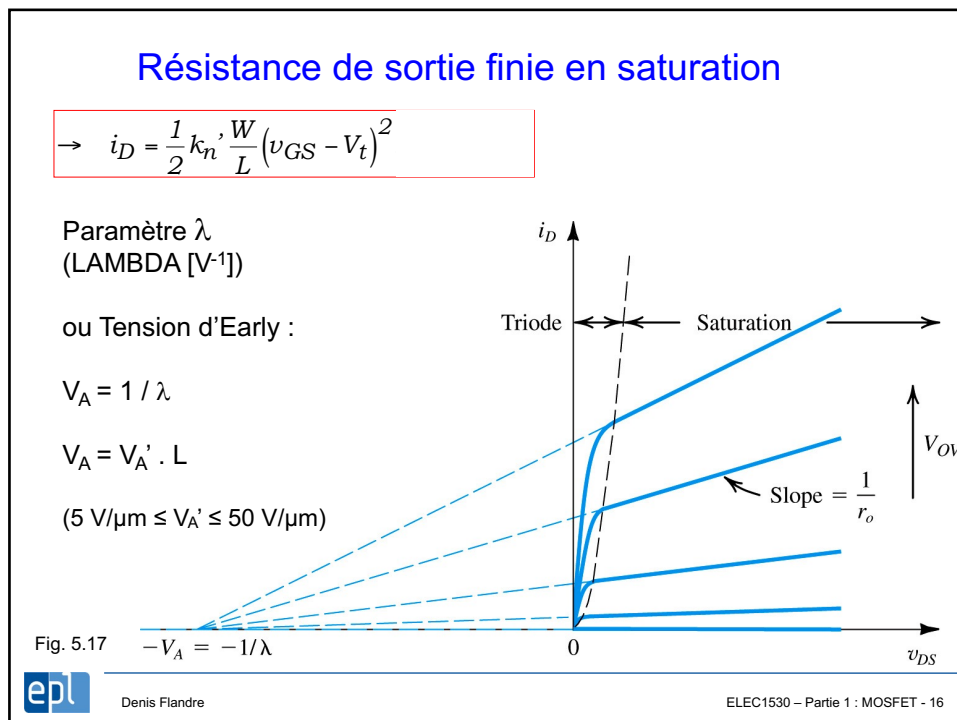


Fig. 5.13

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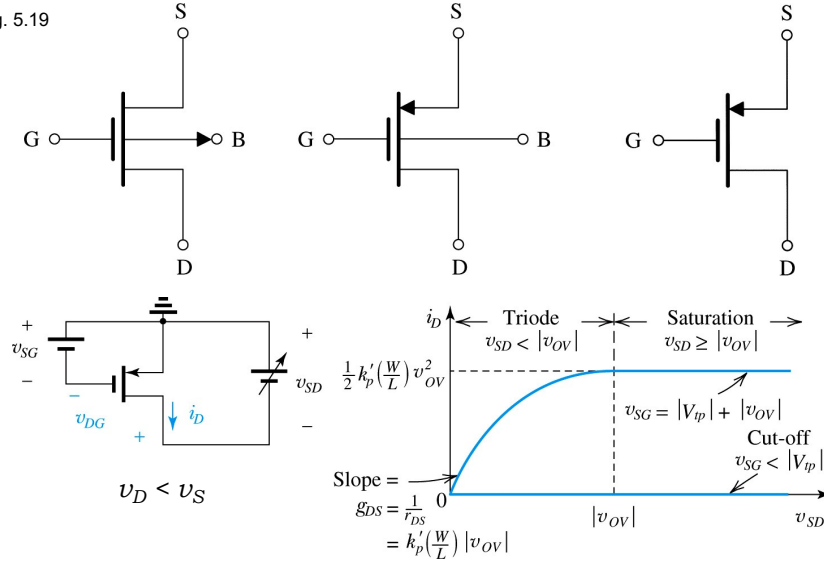
15



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Transistor PMOS

Fig. 5.19



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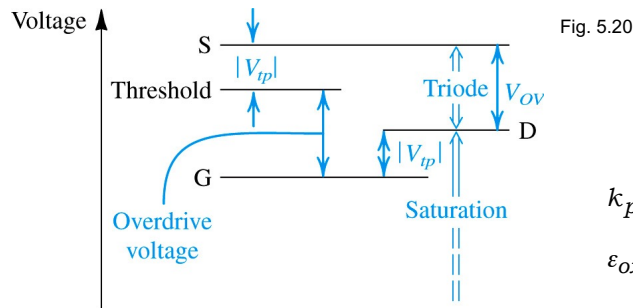


Fig. 5.20

$$k_p' = \mu_p \cdot C_{ox} = \mu_p \frac{\epsilon_{ox}}{t_{ox}}$$

$$\epsilon_{ox} = 3.9 \epsilon_0$$

Saturation Region :

$$v_{SD} \geq |v_{OV}| \Leftrightarrow v_{DG} \leq |V_{tp}|$$

$$i_D = \frac{1}{2} k_p' \frac{W}{L} (v_{SG} - |V_{tp}|)^2 \cdot (1 + |\lambda_p| \cdot v_{SD})$$

Triode Region :

$$v_{SD} < |v_{OV}| \Leftrightarrow v_{DG} > |V_{tp}|$$

$$i_D = k_p' \frac{W}{L} \left[(v_{SG} - |V_{tp}|) v_{SD} - \frac{1}{2} v_{SD}^2 \right]$$

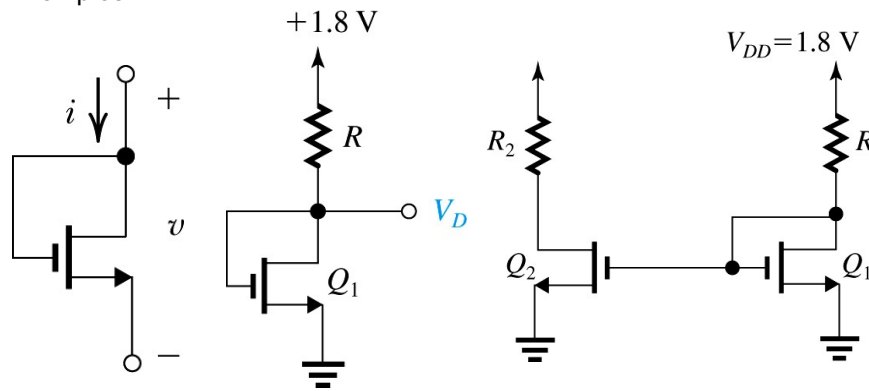
ou

$$i_D = k_p' \frac{W}{L} \left[|v_{OV}| - \frac{1}{2} v_{SD} \right] \cdot v_{SD}$$

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5.3 Circuits MOS en DC

Exemples :



→ TP1 : vendredi



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TP / Labo LTSpice 1 : Polarisation DC

(Re)connaître et savoir écrire / dessiner :

- Symboles des transistors
- Graphiques I_d - V_{ds} pour différents V_{gs}
- Régimes de fonctionnement
- Equations des transistors
- Systèmes d'équations de Kirchoff
- +
- Droite de charge
- Point de fonctionnement DC
- Schéma de polarisation

Remise : le dimanche 04 octobre 2026 à 23h59 au plus tard !

1 fichier à télécharger sur Gradescope : scan du template complété

Noté et partie de la note de l'examen final (cfr. consignes Moodle)



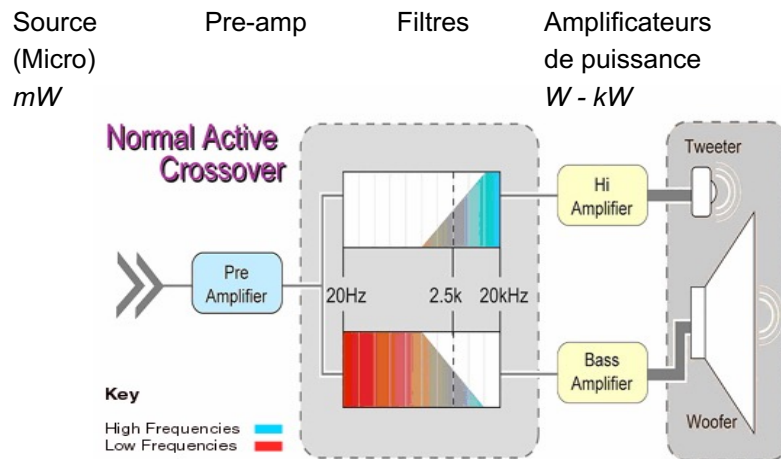
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Amplificateurs

Exemple : Chaîne d'amplification audio



<http://www.moon-audio.com/DEQX.htm14>

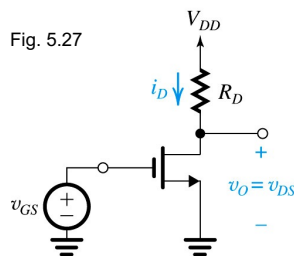


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5.4 Amplificateur MOS en 'grand signal'



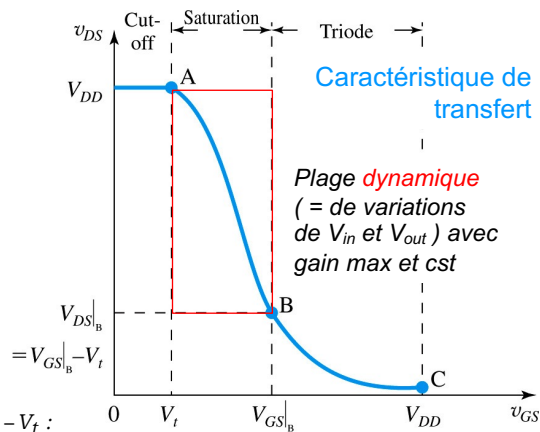
$$v_O = v_{DS} =$$

$$\bullet v_{GS} \leq V_t : i_D = v_O =$$

$$\bullet V_t < v_{GS} \leq V_{GS|_B} \text{ c. - à - d. } v_O > v_{GS} - V_t :$$

$$v_O =$$

$$\Rightarrow V_{GS|_B} =$$



$$\bullet v_{GS} > V_{GS|_B} : \text{triode}$$

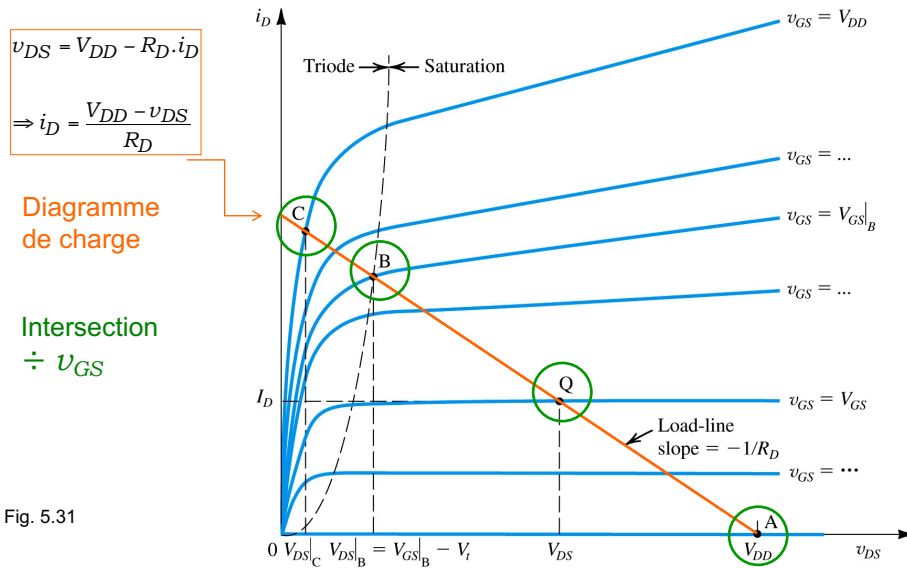


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Analyse graphique – Diagramme de charge

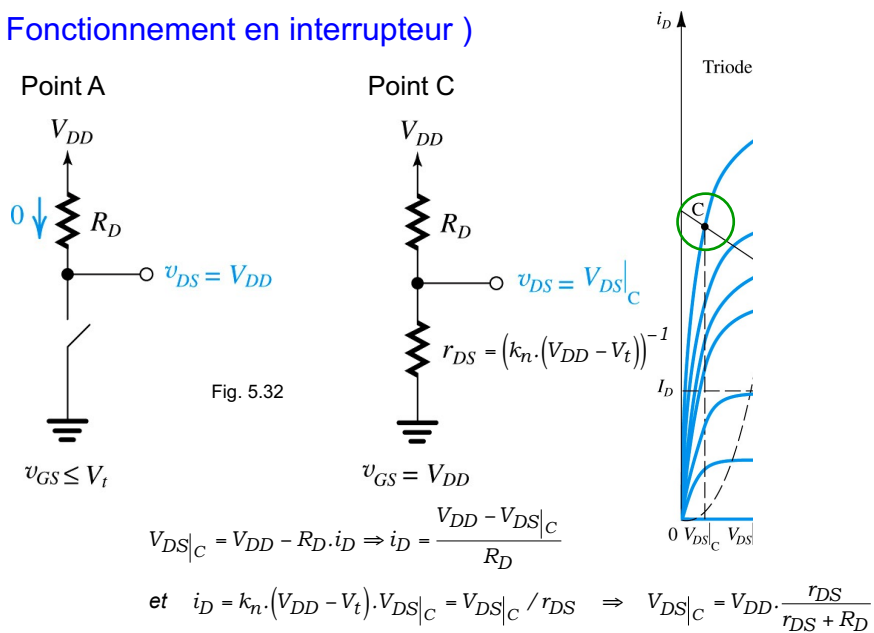


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(Fonctionnement en interrupteur)



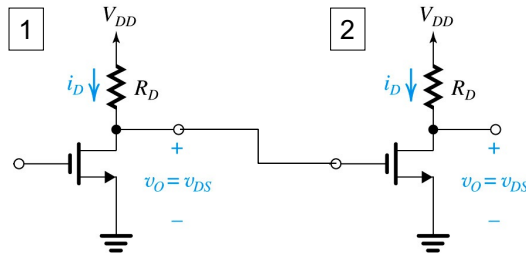
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(Fonctionnement en inverseur)

$$v_{GS1} = V_{DD} \Rightarrow V_{DS}|_C \quad \text{et si } V_{DS}|_C = v_{GS2} < V_t \quad \text{alors} \quad v_{O2} = V_{DD}$$



Inverseur logique :
 $IN = 1 \rightarrow OUT = 0$
 $IN = 0 \rightarrow OUT = 1$

Problèmes de l'inverseur NMOS :
 • $IN = 1 : i_D \neq 0$ et $V_{DS}|_C \neq 0$
 • Gain limité

(Meilleure solution : inverseur CMOS)

Gain

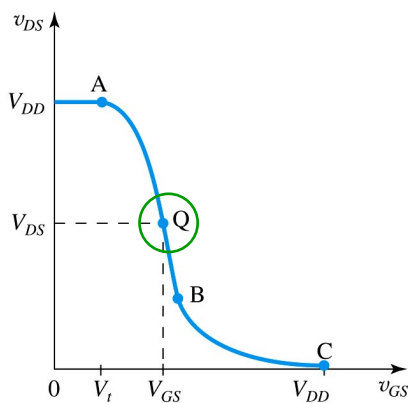


Fig. 5.28

$$A_v(v_{GS}) \equiv \frac{dv_{DS}}{dv_{GS}}$$

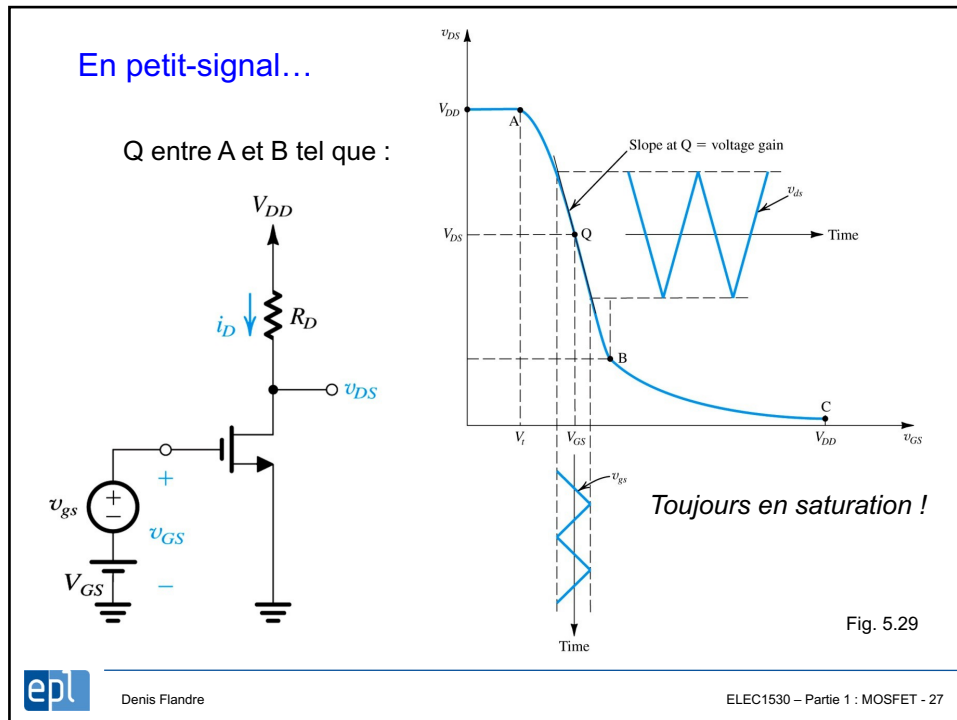
Maximum en saturation

$$v_{DS} = V_{DD} - R_D \cdot \frac{1}{2} k_n (v_{GS} - V_t)^2$$

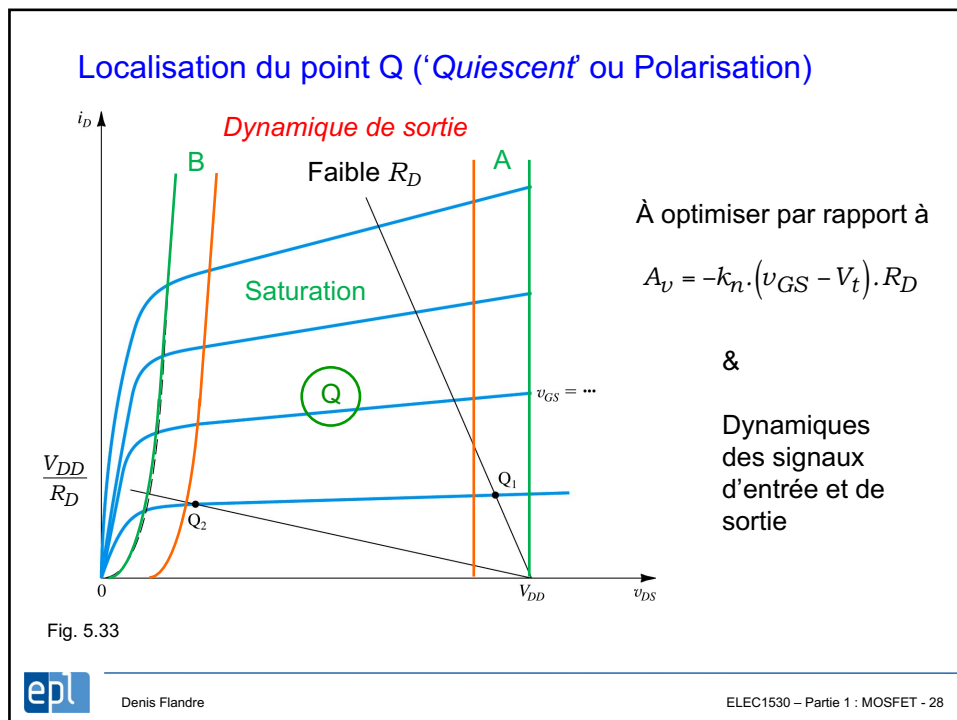
↓

$$A_v = -k_n \cdot (v_{GS} - V_t) \cdot R_D = -k_n \cdot v_{OV} \cdot R_D$$

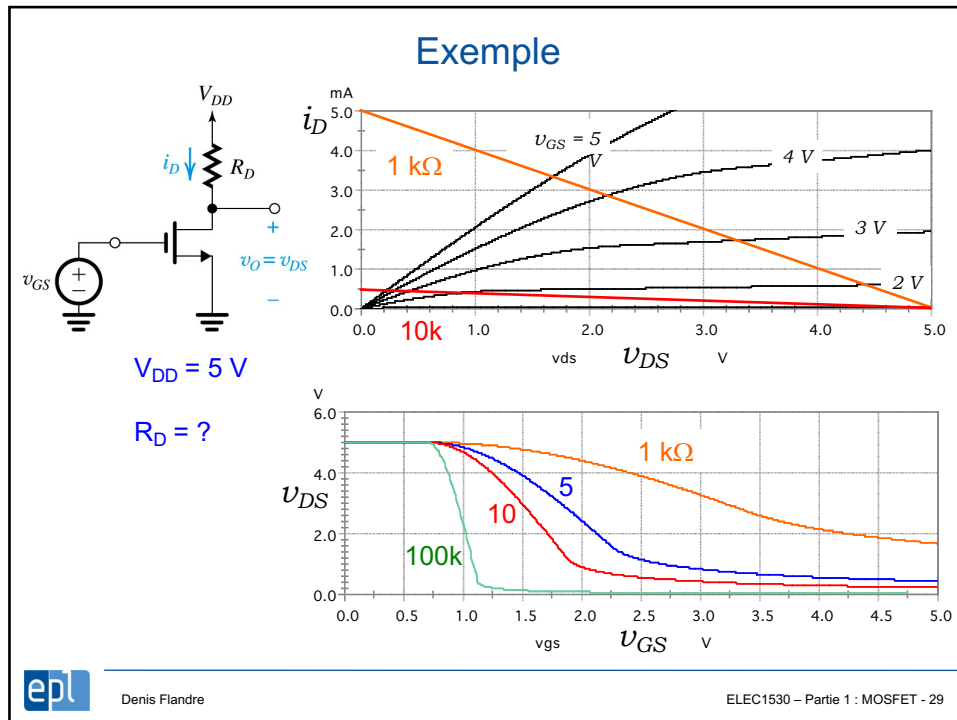
Augmente près de B
mais...



27



28



29

5.5 Fonctionnement et modèles 'petit-signal'

Fig. 5.34

- Point de polarisation DC (.OP)
$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 = \frac{1}{2} k_n V_{OV}^2$$

$$V_{DS} = V_{DD} - R_D \cdot I_D > V_{OV} \text{ (sat.)}$$
- Signal variationnel autour du point OP
$$v_{GS} = V_{GS} + v_{gs} \Rightarrow$$

$$i_D = \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2$$

$$= \underbrace{\frac{1}{2} k_n (V_{GS} - V_t)^2}_{I_D \text{ DC}} + \underbrace{k_n (V_{GS} - V_t) v_{gs}}_{i_d \text{ ac}} + \underbrace{\frac{1}{2} k_n v_{gs}^2}_{\text{distortion non-linéaire}}$$

$$\frac{1}{2} k_n v_{gs}^2 \ll k_n (V_{GS} - V_t) v_{gs}$$

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Approximation petit-signal
pour $v_{gs} \ll 2 \cdot V_{OV}$

$$i_d = k_n (V_{GS} - V_t) v_{gs} = g_m \cdot v_{gs}$$

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t) = k_n V_{OV}$$

Transconductance petit-signal, g_m

Unités : Mhos (Ω^{-1}) ou Siemens (S)

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS}=V_{GS}}$$

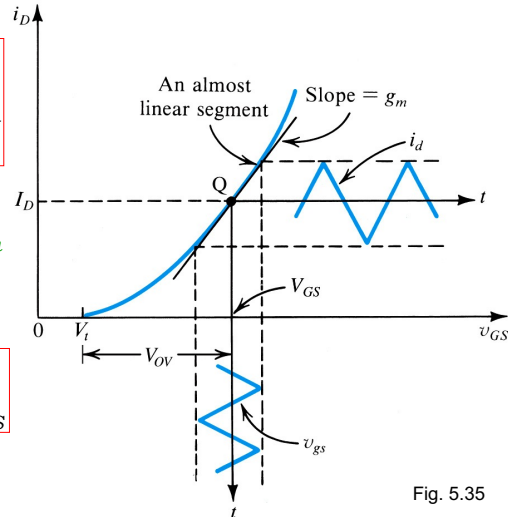


Fig. 5.35

Transconductance petit-signal, g_m

Saturation Region :

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

$$= \frac{1}{2} k_n' \frac{W}{L} V_{OV}^2$$

$$g_m = k_n' \frac{W}{L} (V_{GS} - V_t) = k_n' \frac{W}{L} V_{OV}$$

$$= \sqrt{2 k_n' \frac{W}{L}} \cdot \sqrt{I_D}$$

$$= \frac{2 \cdot I_D}{V_{GS} - V_t} = \frac{2 \cdot I_D}{V_{OV}}$$

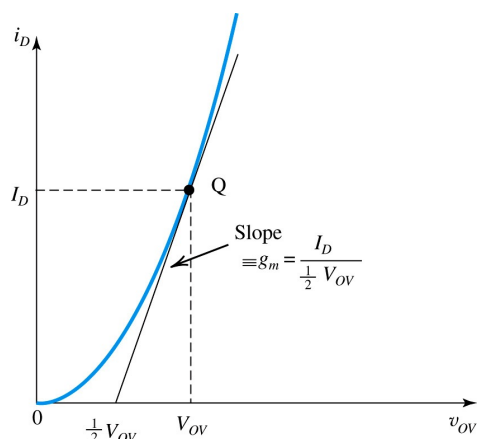
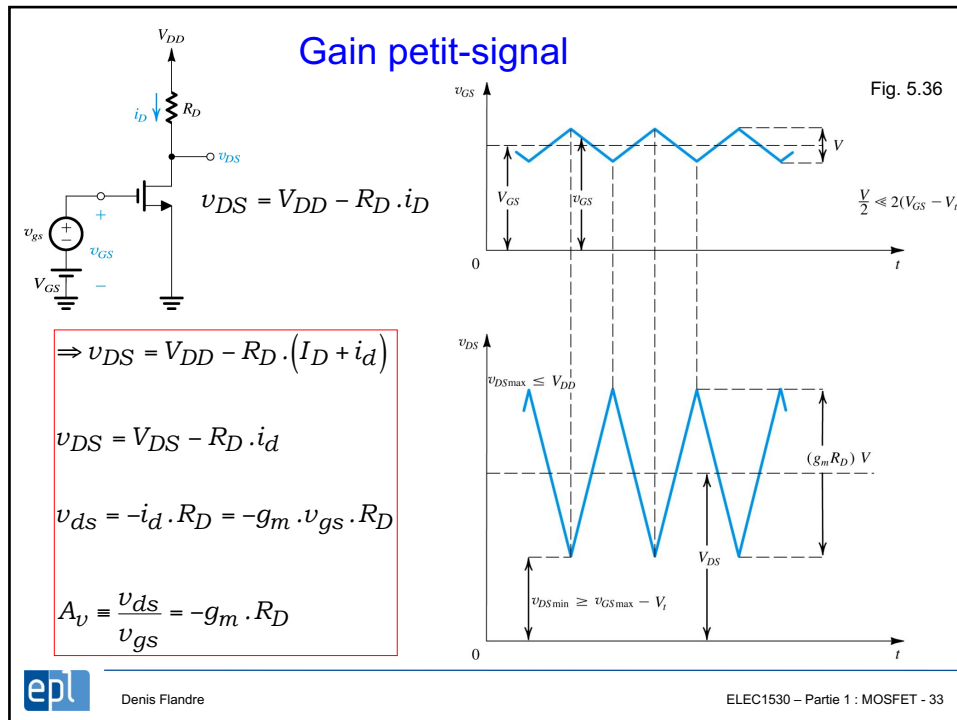
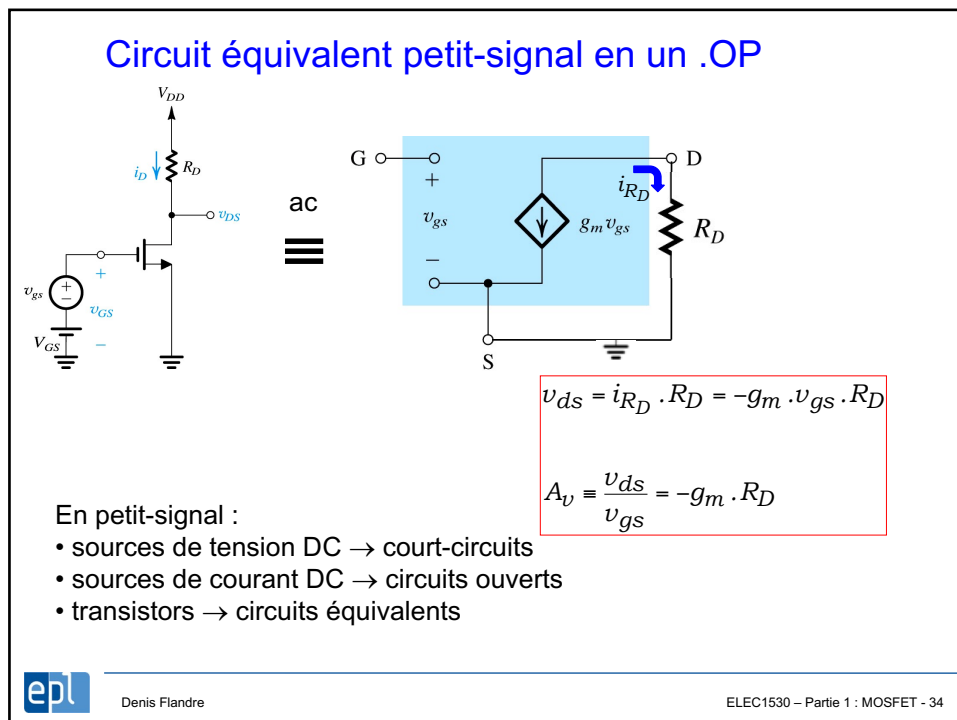


Fig. 5.38



33



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...avec résistance de sortie finie (Effet Early)

$$v_{GS} = V_{GS} + v_{gs} \quad \text{et} \quad v_{DS} = V_{DS} + v_{ds} \Rightarrow$$

$$i_D = \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2 \cdot (1 + \lambda \cdot (V_{DS} + v_{ds}))$$

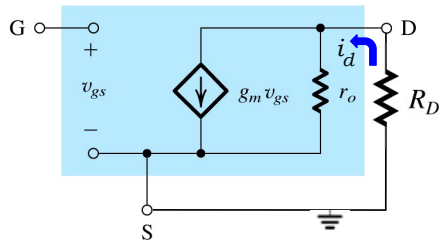
$$\dots \quad i_d = g_m \cdot v_{gs} + g_d \cdot v_{ds}$$

$$g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{V_{GS}, V_{DS}}$$

$$g_d = \left. \frac{\partial i_D}{\partial v_{DS}} \right|_{V_{GS}, V_{DS}}$$

Conductance de sortie petit-signal, g_d

$$r_o = g_d^{-1} = \left(\frac{1}{2} k_n (V_{GS} - V_t)^2 \cdot \lambda \right)^{-1} = \frac{1}{\lambda \cdot I_D} = \frac{V_A}{I_D}$$



$$v_{ds} = -i_D \cdot R_D$$

$$= -(g_m \cdot v_{gs} + v_{ds} / r_o) \cdot R_D$$

$$A_v = \frac{v_{ds}}{v_{gs}} = -g_m \cdot (R_D \parallel r_o)$$

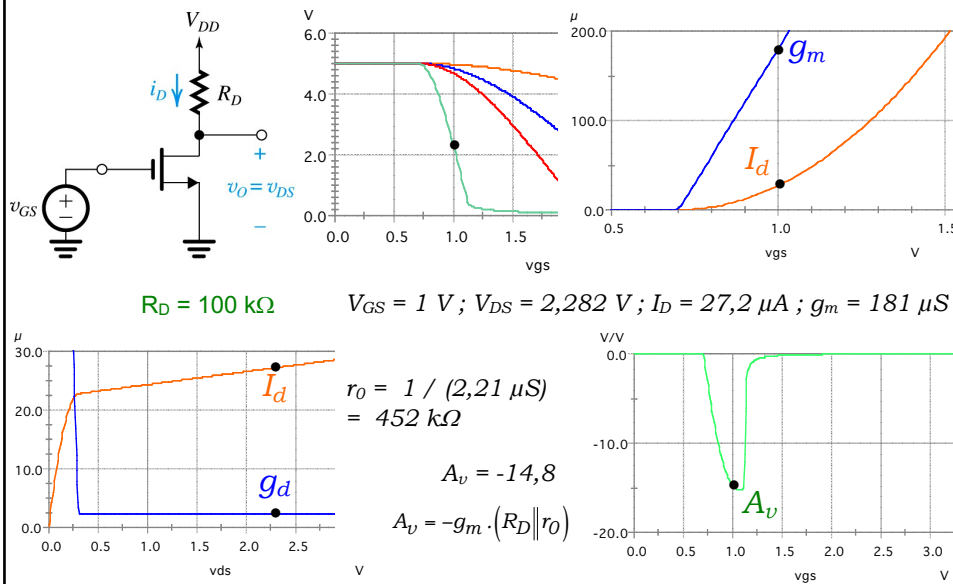


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Exemple : analyse DC vs AC



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Sources DC + ac

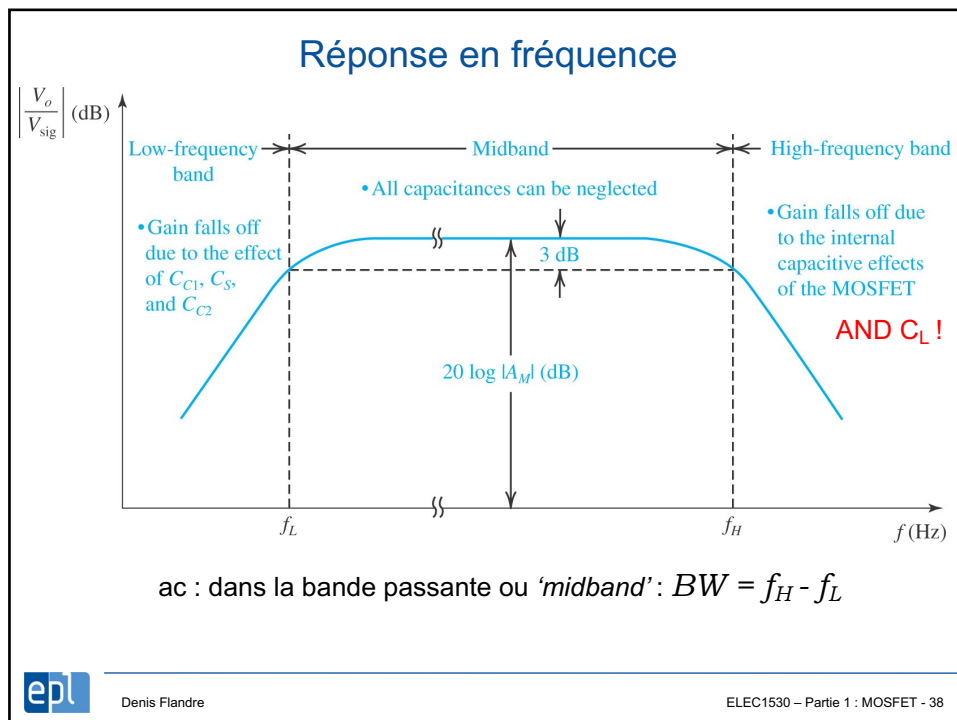
1° Fixer le point de polarisation Q par $R_{G1/2}$ et R_D

2° Appliquer la source petit-signal via capacité de découplage

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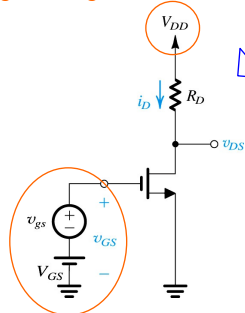


38

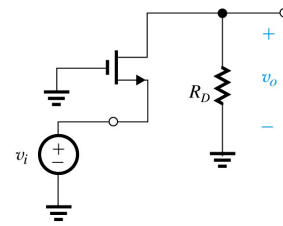
5.6 Configurations d'amplificateurs MOS

3 configurations de base

Représentation
'grand-signal'

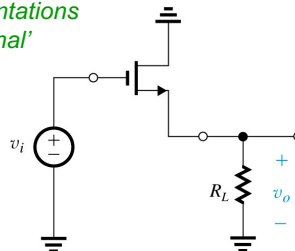


(a) Common Source (CS)



(b) Common Gate (CG)

Représentations
'petit-signal'



(c) Common Drain (CD)

Fig. 5.43

Besoins diversifiés

Exemple : Chaîne d'amplification audio

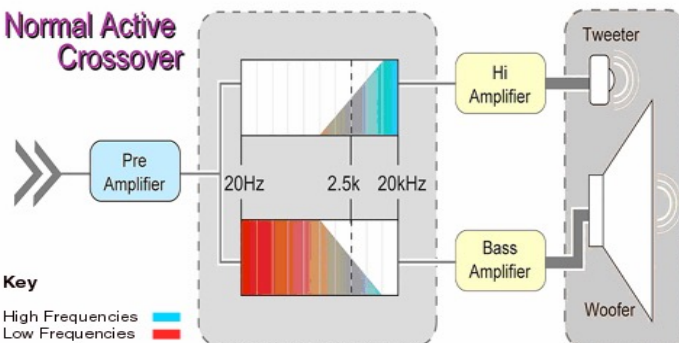
Source
(Micro)
 $10 \dots k\Omega$

Pre-amp
 $R_{in} \approx M\Omega$

Filtres

Amplificateurs
de puissance
 $R_{out} \approx \Omega$

Normal Active
Crossover



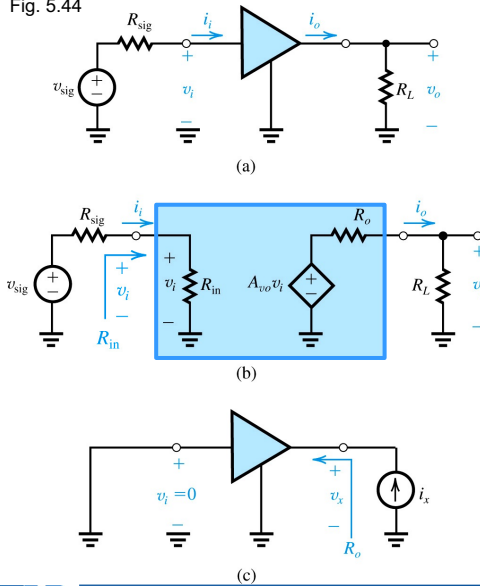
Key

High Frequencies
Low Frequencies

<http://www.moon-audio.com/DEQX.htm14>

Caractérisation des amplificateurs (ac, bp)

Fig. 5.44



$$R_{in} = \frac{v_i}{i_i}$$

$$v_i = \frac{R_{in}}{R_{in} + R_{sig}} \cdot v_{sig}$$

$$A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L = \infty}$$

$$R_o = \left. \frac{v_x}{i_x} \right|_{v_i = 0}$$

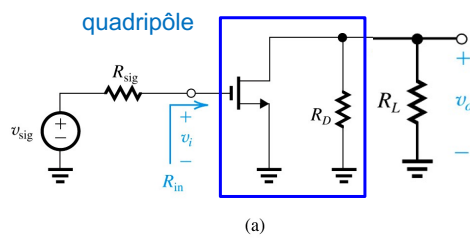
$$v_o = \frac{R_L}{R_L + R_o} \cdot A_{vo} \cdot v_i$$

$$A_v = \frac{v_o}{v_i} = A_{vo} \cdot \frac{R_L}{R_L + R_o}$$

$$G_v = \frac{v_o}{v_{sig}}$$

$$G_v = \frac{R_{in}}{R_{in} + R_{sig}} \cdot A_{vo} \cdot \frac{R_L}{R_L + R_o}$$

Amplificateur en source commune (ac)



$$R_{in} =$$

$$R_o =$$

$$v_o =$$

$$A_{vo} =$$

$$v_i =$$

$$A_v =$$

$$G_v =$$

Fig. 5.45

(b)

... avec résistance de source ('*degeneration*')

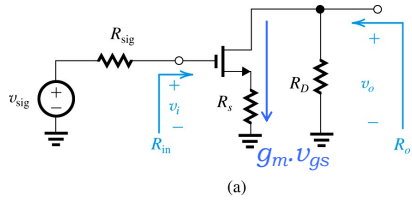
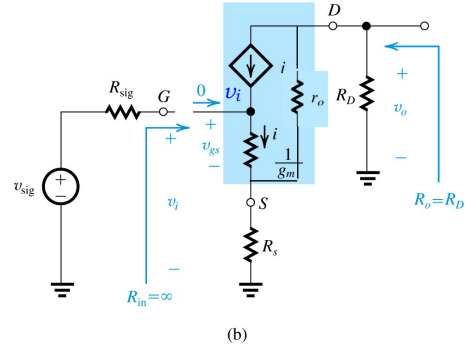


Fig. 5.47



En négligeant g_d :

$$R_{in} = \infty$$

$$v_{gs} = \left(g_m^{-1} / g_m^{-1} + R_S \right) \cdot v_i$$

$$= v_i / 1 + g_m \cdot R_S$$

$$v_o = -i.R_D$$

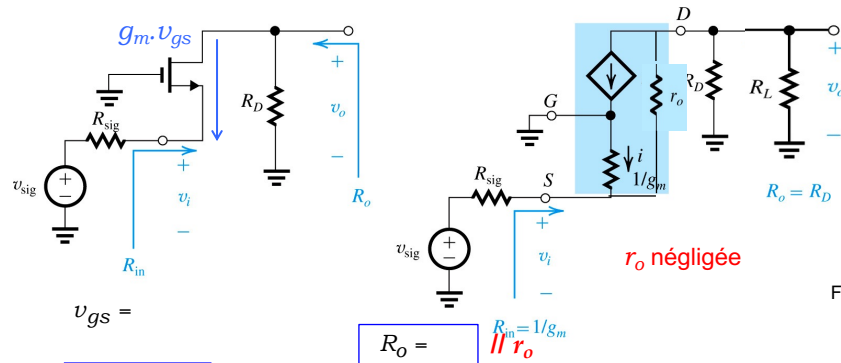
$$A_{v_o} = \frac{v_o}{v_i} = -\frac{R_D}{g_m^{-1} + R_S} = -\frac{g_m \cdot R_D}{1 + g_m \cdot R_S}$$

...

= "negative feedback" (Chap.10)

⇒ Amélioration de performances
au prix d'un gain plus faible !

Amplificateur en grille commune (ac)



r_o négligée

Fig. 5.48

$$v_{gs} =$$

$$R_{in} =$$

$$v_O =$$

$$i =$$

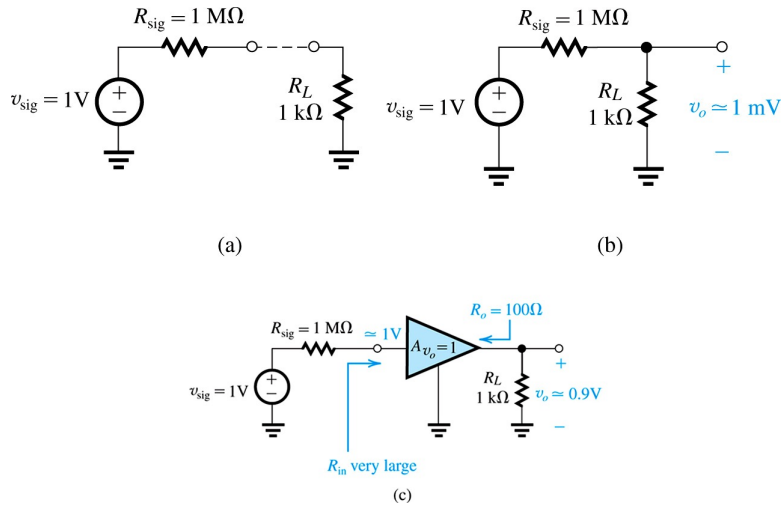
$$A_{vo} \equiv \frac{v_o}{v_i} =$$

$$\frac{v_i}{v_{sig}} =$$

$$G_v \equiv \frac{v_o}{v_i} \frac{v_i}{v_{sig}} =$$

Amplificateur en drain commun

Source suiveur ou Buffer, par ex. pour adaptation d'impédance

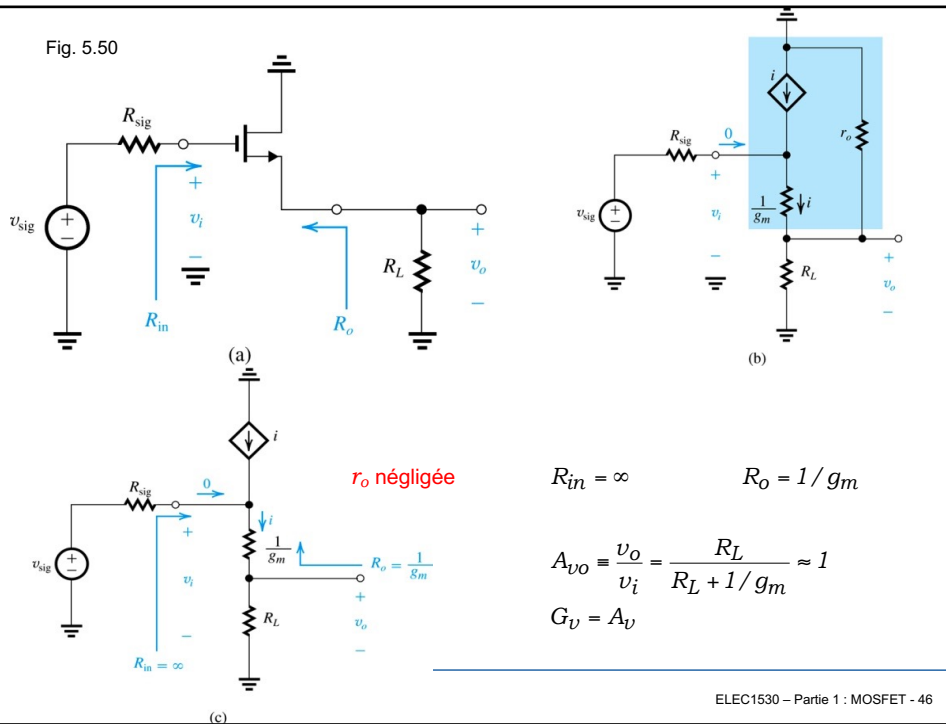


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Fig. 5.50



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En résumé

- CS : étage de gain le plus classique (*seul ou en cascade*)
- CS + R_s : « *linéarisation* »
- CG : $R_{in} = 1 / g_m$, mais même gain que CS + R_s
c.-à-d. = $(R_{tot} \text{ vue au D}) / (R_{tot} \text{ vue à la S})$
(*applications HF quand R_{sig} faible, par ex. 50 Ω*)
- CD : $R_o = 1 / g_m$ et $A_{vo} \approx 1$
(*applications à faible R_L ou suiveur de tension*)

5.7 Polarisation des amplificateurs MOS

Fixer V_G et I_D dans le transistor $I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$

÷ technologie, dimensions et spécifications de l'application

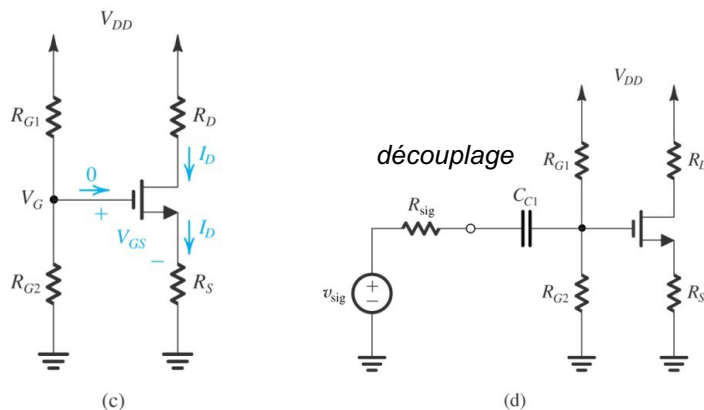
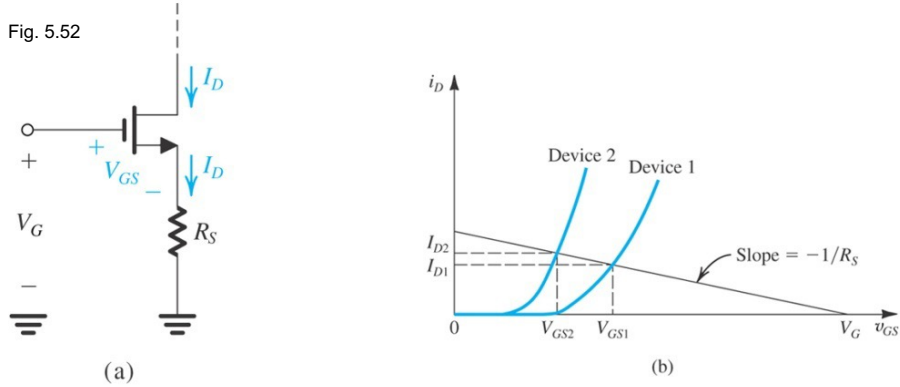


Fig. 5.52

Polarisation stable (par rapport aux variations de la technologie, de la température et de la tension d'alimentation)

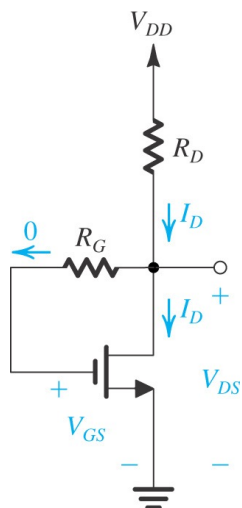
Fig. 5.52



$$V_G = V_{GS} + R_S \cdot I_D$$

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 \quad \text{'feedback'}$$

... ou



Rem.: Analyse DC, ici
4 inconnues (I_D , I_G , V_G , V_D)
 $\Rightarrow$ 4 équations
+ 2 conditions de saturation

$$R_G \text{ [M}\Omega\text{]}$$

$$I_G = 0$$

$$V_{GS} = V_{DS} = V_{DD} - R_D \cdot I_D$$

$$V_{DD} = V_{GS} + R_D \cdot I_D$$

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2$$

'stabilisation'

Fig. 5.54

Polarisation par source de courant

Amplificateur (CS)

et son circuit de polarisation (miroir de courant)

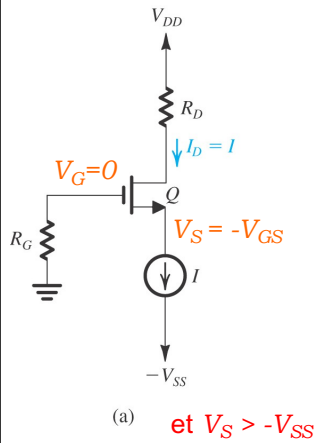
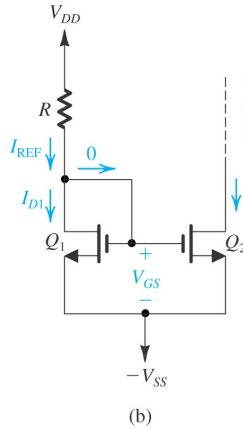


Fig. 5.55



To source of transistor Q in Fig. 5.55 (a)

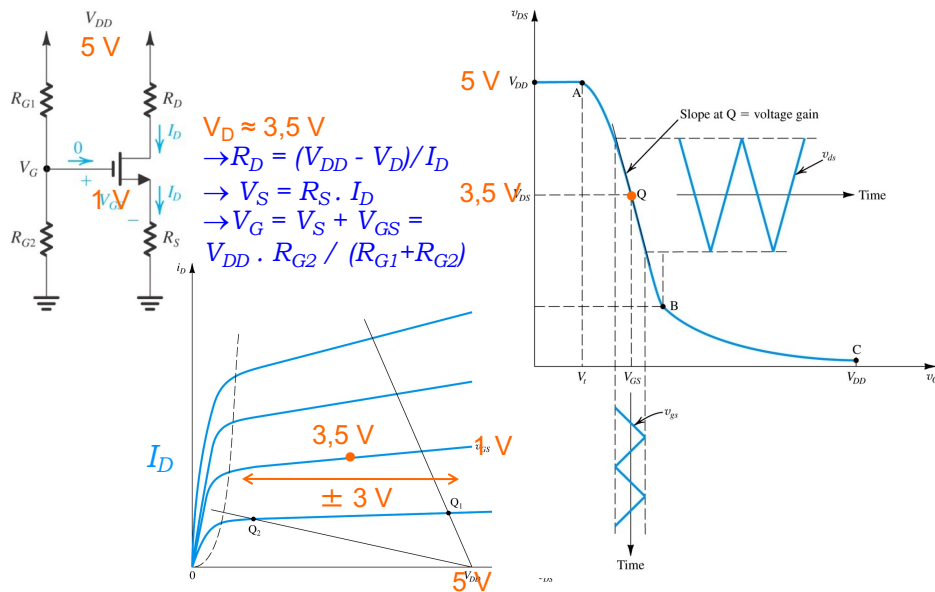
$$I_{D1} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_1 (V_{GS} - V_t)^2$$

$$I_{D1} = I_{REF} = \frac{V_{DD} + V_{SS} - V_{GS}}{R}$$

$$I = I_{D2} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_2 (V_{GS} - V_t)^2$$

$$I = I_{REF} \cdot \frac{\left(\frac{W}{L} \right)_2}{\left(\frac{W}{L} \right)_1}$$

Exemple



5.8 Amplificateurs MOS discrets

Composants : AO + R, C extérieures ('off-the-shelf') sur plaquette ou PCB

≠ AO intégrés complètement sur un chip de Si (r_o pas négligeable)

⇒ Configuration (5.6) + Polarisation DC (5.7) + (Dé)couplage ac (9)

Transistor nMOS
Source commune

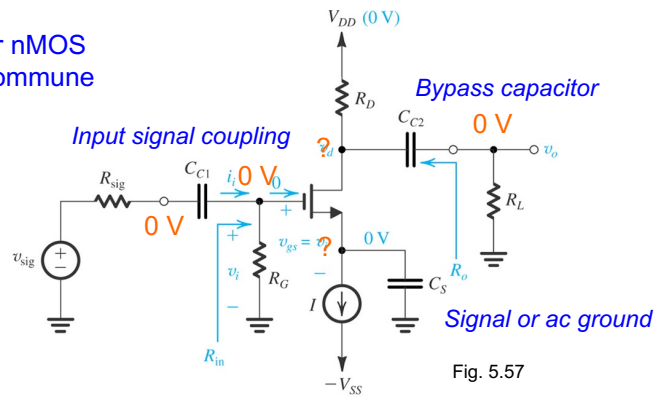


Fig. 5.57

DC ($C = \text{circuits ouverts, impédance } \infty$)

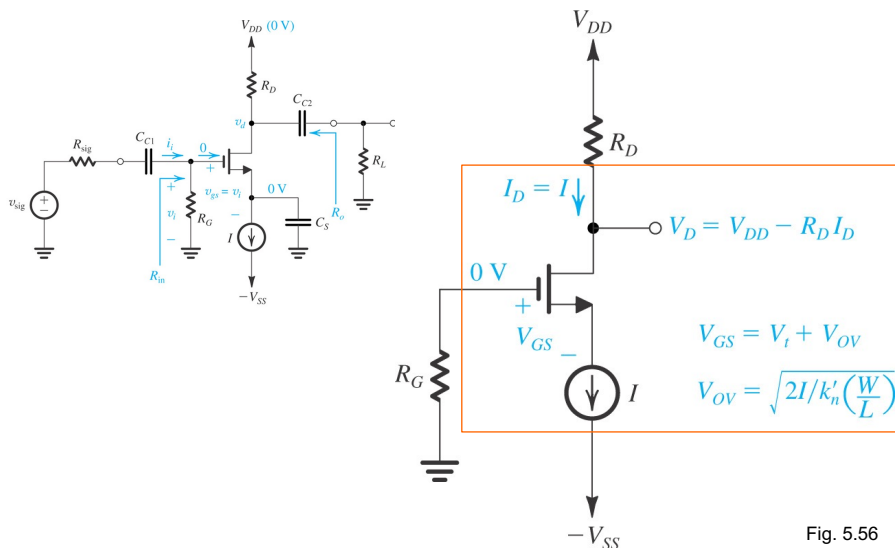
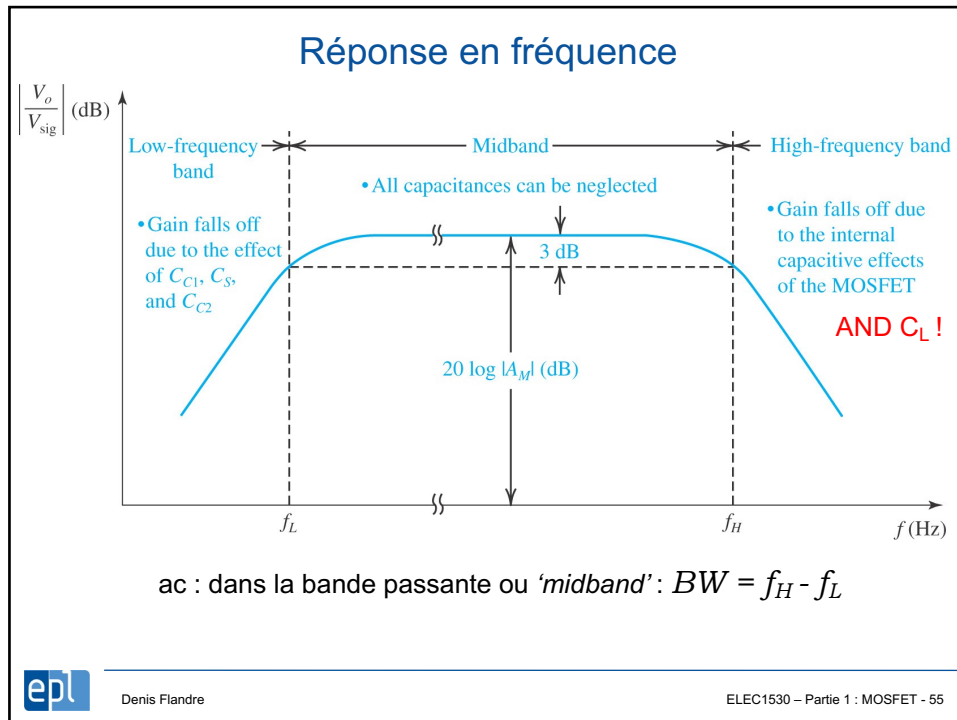
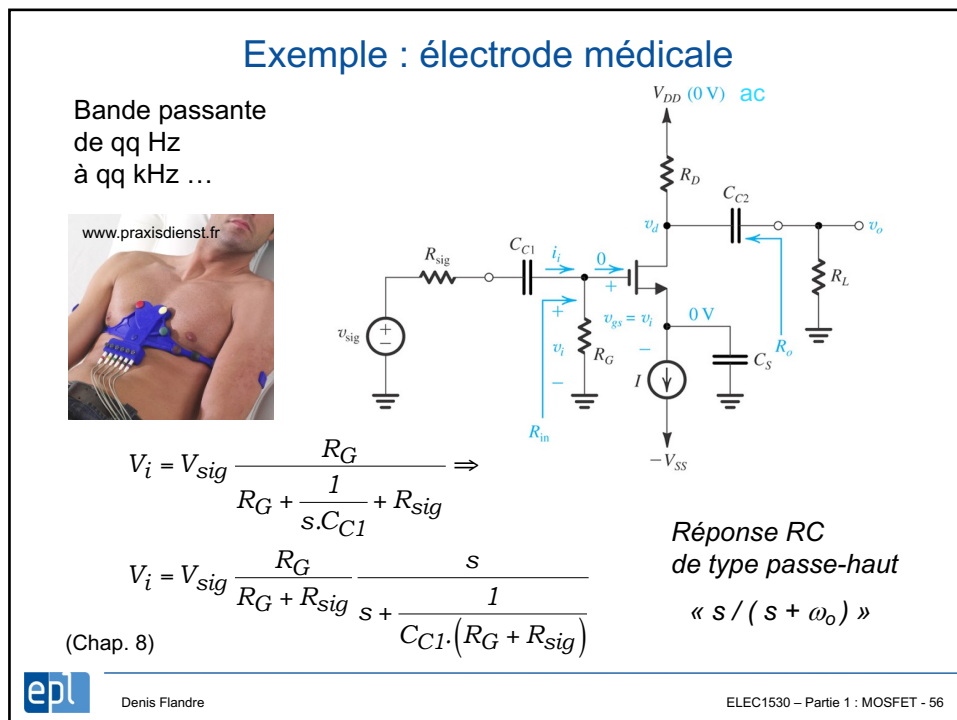


Fig. 5.56



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AC (dans la bande passante, C découplage = circuits fermés,
V alim = masse ac, source I = circuit ouvert)

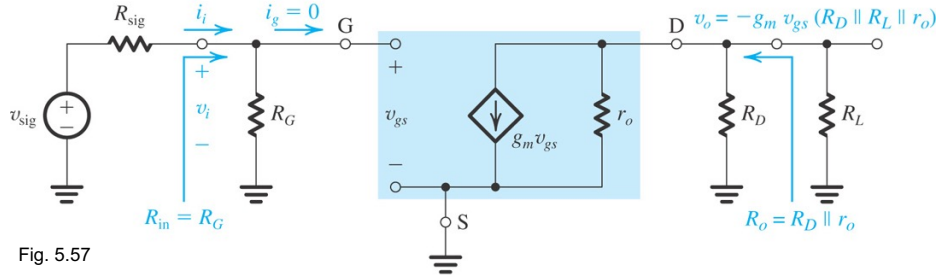
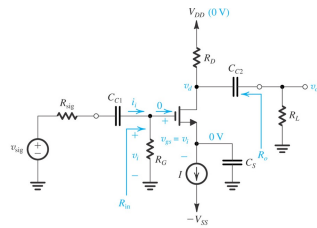


Fig. 5.57

Cas CS



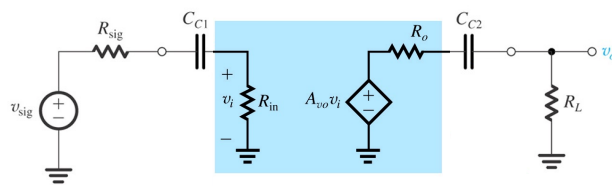
(b)

$$R_{in} = R_G \quad R_o = R_D \parallel r_o$$

$$A_{vo} = -g_m \cdot (R_D \parallel r_o)$$

$$G_v = -\frac{R_G}{R_G + R_{sig}} g_m \cdot (R_D \parallel R_L \parallel r_o)$$

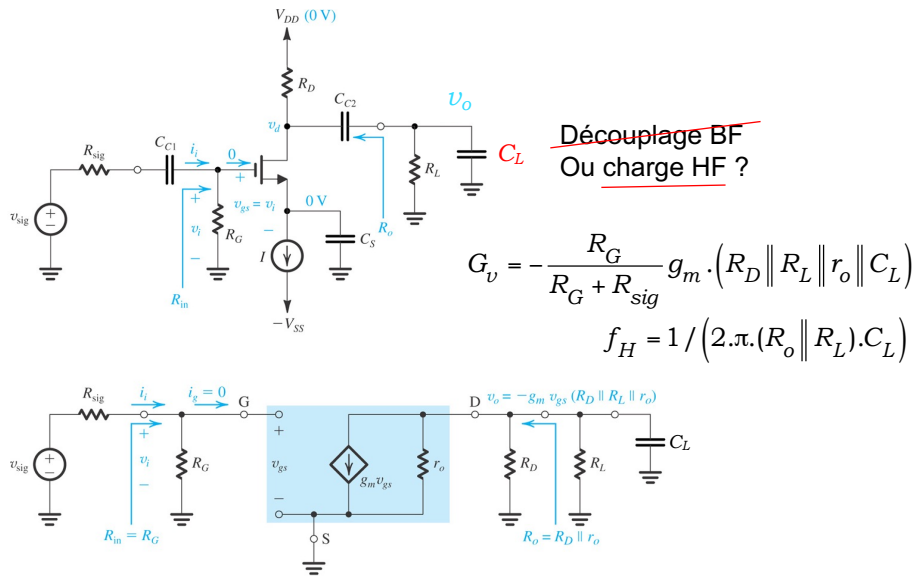
Estimation rapide des pôles basse fréquence (f_L)



$$R_{in} = R_G \quad v_i = v_{sig} \frac{R_{in}}{R_{in} + R_{sig}}$$

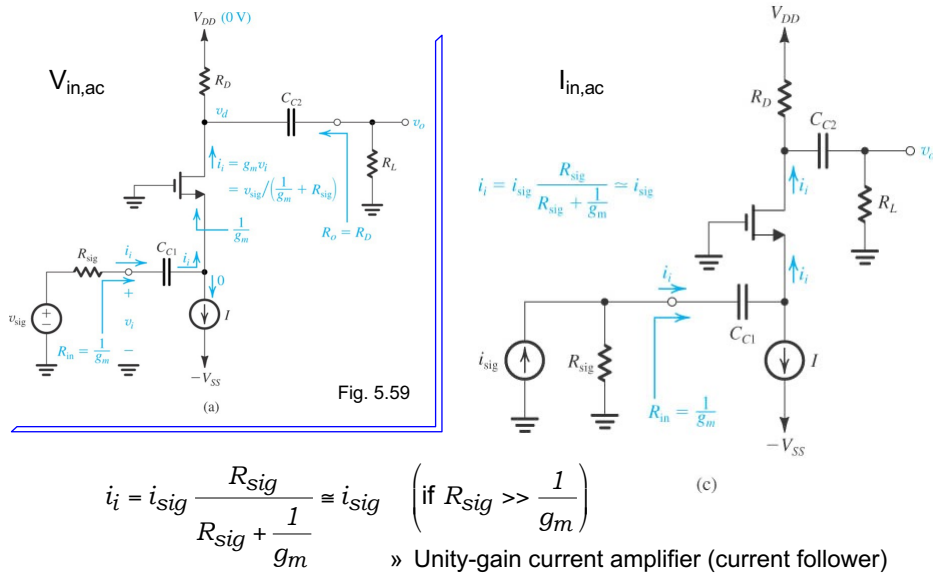
$$R_o = R_D \parallel r_o \quad v_o = A_{vo} \cdot v_i \frac{R_L}{R_L + R_o}$$

Et à haute fréquence (f_H) ?



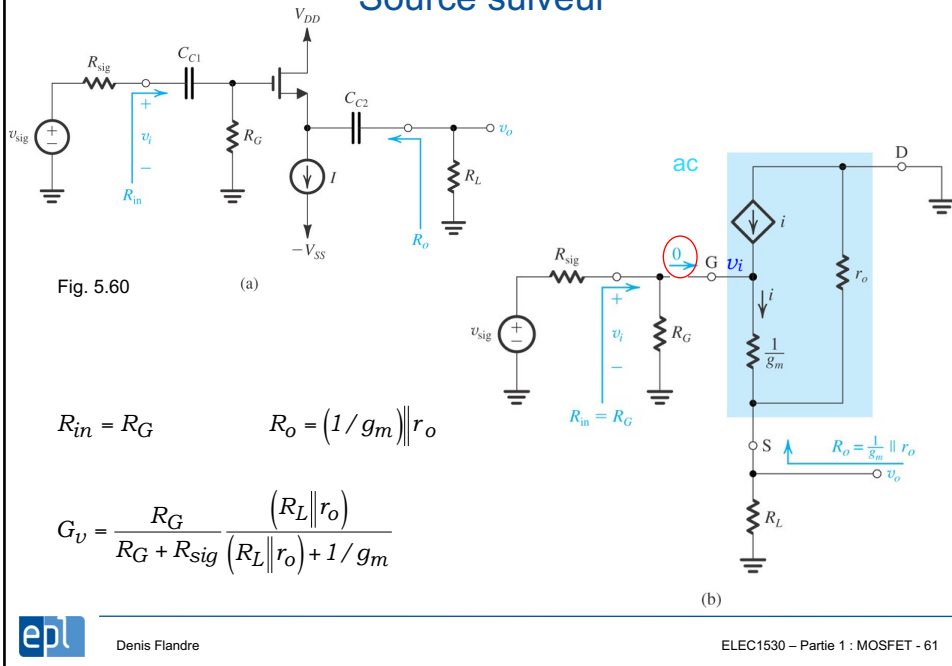
59

Grille commune



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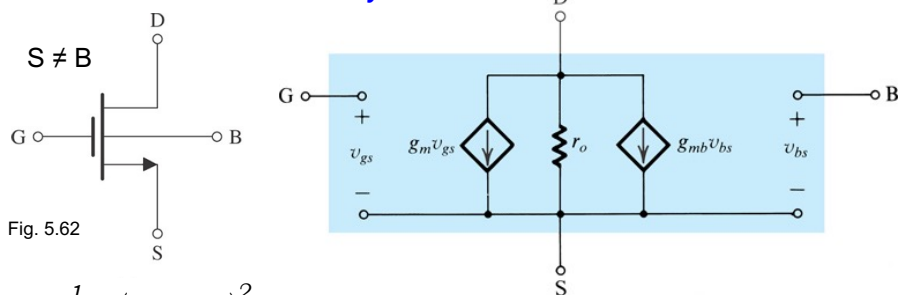
Source suiveur



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5.9 Effets supplémentaires

Effet du substrat ou 'Body effect'



$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2$$

$$V_t = V_{t0} + \gamma \cdot [\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}]$$

γ : facteur technologique (GAMMA)
paramètre d'effet de substrat

$$g_{mb} = \frac{\partial i_D}{\partial v_{BS}} \bigg|_{v_{GS}=cst, v_{DS}=cst} = \chi \cdot g_m$$

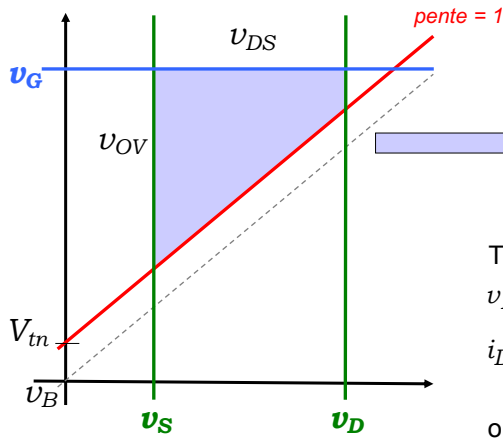
$$\chi = \frac{\partial V_t}{\partial v_{SB}} = \frac{\gamma}{2\sqrt{2\phi_f + V_{SB}}}$$

(pente > 1 dans diagramme de Jaspers)

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Diagramme de Jaspers

courant en régime non saturé



Surface du trapèze

$$= i_D / \left(k_n' \frac{W}{L} \right)$$

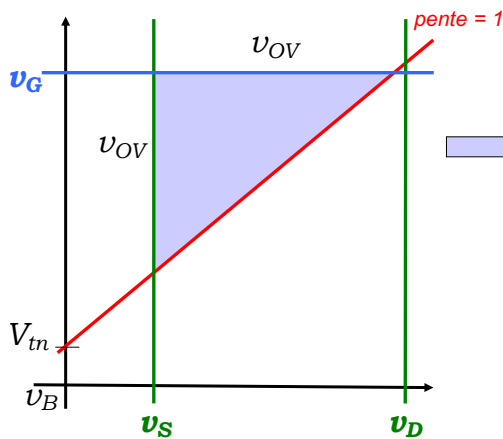
Triode Region :

$$v_{DS} < v_{OV} \Leftrightarrow v_{GD} > V_{tn}$$

$$i_D = k_n' \frac{W}{L} \left[(v_{GS} - V_{tn}) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

$$\text{ou } i_D = k_n' \frac{W}{L} \left[v_{OV} - \frac{1}{2} v_{DS} \right] \cdot v_{DS}$$

courant en régime saturé



Surface du triangle

$$= i_D / \left(k_n' \frac{W}{L} \right)$$

Saturation Region :

$$v_{DS} \geq v_{OV} \Leftrightarrow v_{GD} \leq V_{tn}$$

$$\rightarrow i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2$$

$$= \frac{1}{2} k_n' \frac{W}{L} v_{OV}^2$$

Paramètres intrinsèques ('channel region') et extrinsèques (parasites)

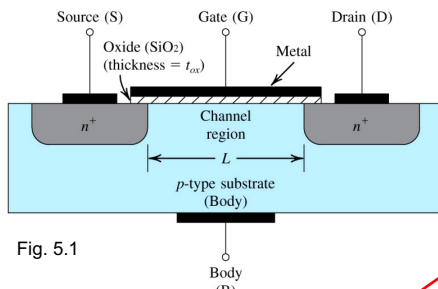
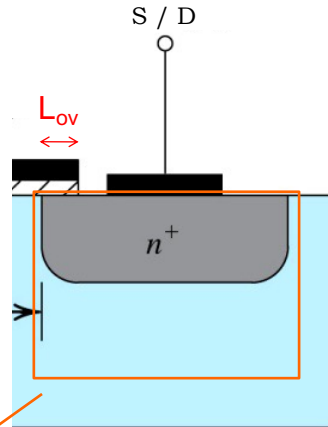


Fig. 5.1

Recouvrement ('overlap') : LD
 $\Rightarrow L' = \text{longueur effective du canal}$
 $= L_G - 2 \cdot L_{ov} !!!$
 Et capacités additionnelles...



Diodes de source / drain – body : Js, CJ, AS, AD, PS, PD...

$$AD = AS = 2.75L_{\min}W \quad PD = PS = 2 \times 2.75L_{\min} + W$$



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Table B.2 Parameters of the SPICE Level-1 MOSFET Model (Partial Listing)

SPICE Parameter	Book Symbol	Description	Units
LEVEL		Basic Model Parameters MOSFET model selector	
TOX	t_{ox}	Gate-oxide thickness	m
COX	C_{ox}	Gate-oxide capacitance, per unit area	F/m ²
UO	μ	Carrier mobility	cm ² /V s
KP	k'	Process transconductance parameter	A/V ²
LAMBDA	λ	Channel-length modulation coefficient	V ⁻¹
VTO	V_{t0}	Threshold Voltage Parameters Zero-bias threshold voltage	V
GAMMA	γ	Body-effect parameter	V ^{1/2}
NSUB	N_A, N_D	Substrate doping	cm ⁻³
PHI	$2\Phi_f$	Surface inversion potential	V
JS		MOSFET Diode Parameters Body-junction saturation-current density	A/m ²
CJ		Zero-bias body-junction capacitance, per unit area over the drain/source region	F/m ²
MJ		Grading coefficient, for area component	
CJSW		Zero-bias body-junction capacitance, per unit length along F/m the sidewall (periphery) of the drain/source region	F/m
MJSW		Grading coefficient, for sidewall component	
PB	V_b	Body-junction built-in potential	V
LD	L_{ov}	MOSFET Dimension Parameters Lateral diffusion into the channel from the source/drain diffusion regions	m
WD		Sideways diffusion into the channel from the body along the width	m
CGBO		MOS Gate-Capacitance Parameters Gate-body overlap capacitance, per unit channel length	F/m
CGDO	C_{ov}/W	Gate-drain overlap capacitance, per unit channel width	F/m
CGSO	C_{ov}/W	Gate-source overlap capacitance, per unit channel width	F/m

Attention par abus de langage :

$$L = L' = L_{eff}$$

$$W = W' = W_{eff}$$

par la suite (équations)

MAIS SPICE :

$$M \times 1 \ 2 \ 3 \ 4 \ L=y \ W=z$$

$$L_{eff} = L - 2LD$$

$$W_{eff} = W - 2WD$$

$$I_D \div L_{eff}, W_{eff}$$

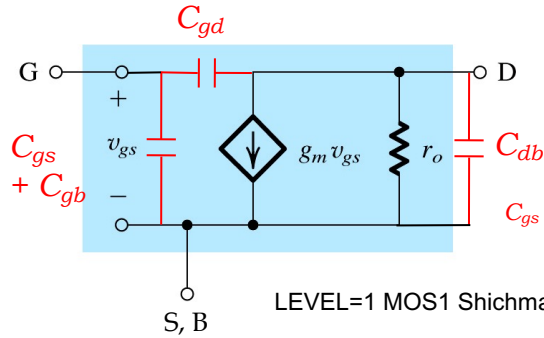


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Modèle petit-signal complet : capacités



Transistor ON
en saturation :

$$C_{gs} = 2/3 \cdot (C_{ox} \cdot W_{eff} \cdot L_{eff}) + W \cdot CGSO$$

$$C_{gd} = W \cdot CGDO$$

$$C_{gb} = L \cdot CGBO$$

(Chap. 9)

$$C_{db} = \frac{CJ}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJ}} AD + \frac{CJSW}{\left(1 + \frac{V_{DB}}{PB}\right)^{MJSW}} PD$$

$$C_{sb} = \frac{CJ}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJ}} AS + \frac{CJSW}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJSW}} PS$$



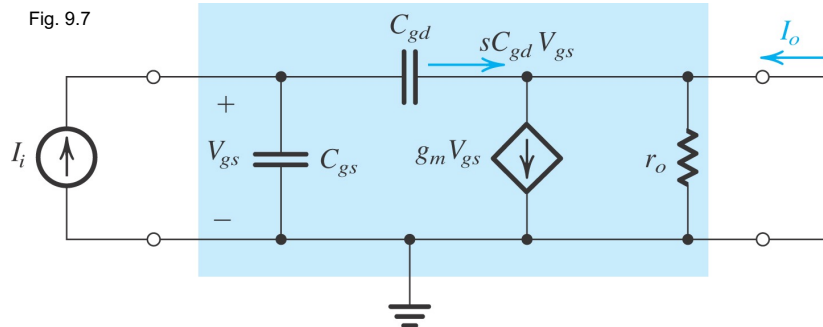
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Limite haute fréquence intrinsèque

Fig. 9.7



$$I_o = g_m \cdot V_{gs} - s \cdot C_{gd} \cdot V_{gs} \quad V_{gs} = I_i / s(C_{gs} + C_{gd})$$

$$\frac{I_o}{I_i} \approx \frac{g_m}{s(C_{gs} + C_{gd})} = 1 \Rightarrow \omega_T = \frac{g_m}{C_{gs} + C_{gd}} = 2\pi \cdot f_T$$

$$f_T : 100 \text{ MHz } (5 \mu\text{m}) \dots > 100 \text{ GHz } (< 100 \text{ nm})$$

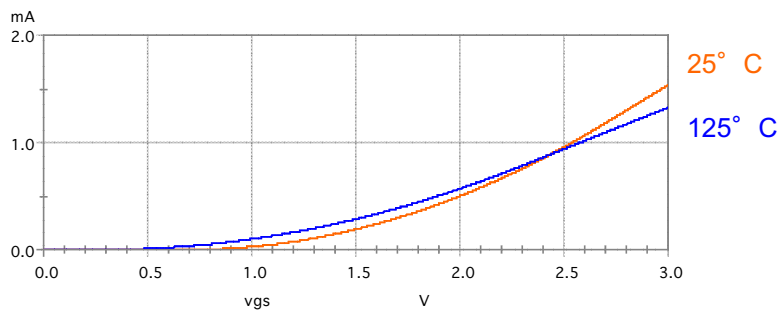


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Effet de la température



$$V_t(T) = V_t(25^\circ \text{C}) - TC \cdot (T - 25^\circ \text{C})$$

$$TC \approx 2 - 3 \text{ mV/}^\circ \text{C}$$

(attention : 'SPICE model level 1' ?)

$$k_n'$$

$$\mu_0(T) = \frac{\mu_0(T_0)}{\left[\frac{T}{T_0} \right]^{1.5}}$$



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Transistor MOS en dépletion

Cas particulier : $V_{tn} < 0$ (NMOS)

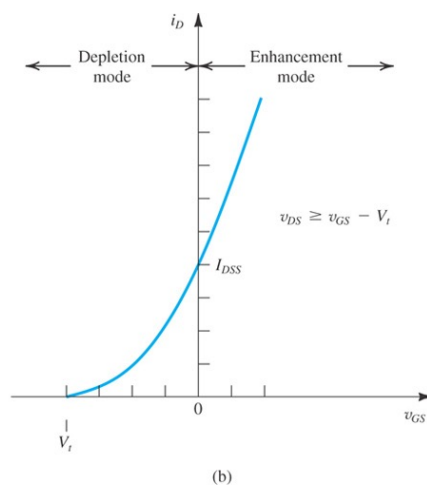
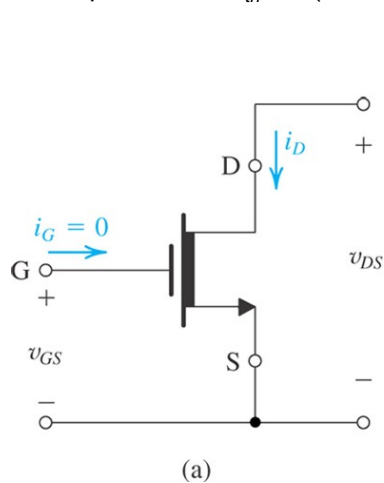


Fig. 5.63



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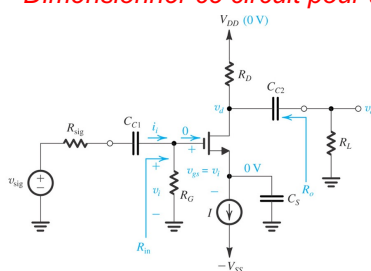
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Contenu du chapitre

- 5.1 Structure du transistor MOS : *Paramètres du modèle SPICE*
- 5.2 Caractéristiques courant – tension :
 - *Régimes de fonctionnement, Equations I-V*
- 5.3 Circuits MOS DC
- 5.4 Amplificateurs MOS en 'grand-signal'
 - *Caractéristique de transfert, Dynamiques des signaux, Gain, Inverseur*
- 5.5 Fonctionnement et modèles 'petit-signal'
 - *Transconductance, Résistance de sortie, Gain, Circuit équivalent*
- 5.6 Configurations d'amplificateurs MOS
 - *CS, CS + R_s , CG, CD (suiveur)*
- 5.7 Polarisation : *par V_G , R_S , R_G ou source de I*
- 5.8 Amplificateurs MOS discrets : *DC, AC*
- 5.9 Effets supplémentaires : *substrat, capacités, température*

Méthodologie (≈ question examen)

Dimensionner ce circuit pour une application avec un transistor donné



- Données : MOS (I-V), R_{sig} , R_L , G_v , bande passante
- Inconnues : I_{bias} , R_G , R_D , C

⇒ **Système d'équations interdépendantes :**

- DC : $\forall$ nœuds : KVL, KCL, $V = R \cdot I$ & équations du transistor

$$\bullet \text{ AC : } G_v = -\frac{R_G}{R_G + R_{sig}} g_m \cdot (R_D \parallel R_L \parallel r_o) \cdot \frac{s}{s + \frac{1}{C_{C1} \cdot (R_G + R_{sig})}}$$

$$\Rightarrow R_G \gg R_{sig}, R_o \ll R_L \text{ \& } (g_m, R_D, I_{bias}), C_{C1} \dots$$

'Exercices' théoriques (= *matière de l'examen*)

- Analyser le point DC (tous les V et I) de tous les schémas des slides ...
- Calculer R_{in} , R_{out} , A_v de l'étage CG avec r_0 .
- Idem avec autres polarisations comme slides 48-51.
- Idem Source suiveur (slide 61)
- Etc...
- Solutions dans Sedra-Smith



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Travail LT - Spice 2

- Cfr. Moodle



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